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Volume 04 — M36–M47

Information Security (CEC) · SWAYAM / CEC Information Security **notes-3** — concept exam notes (definitions & what things do). Breadth over lab depth.

M36: Introduction to Mobile IP and Addressing

Learning focus

- Explain why classic IP prefixes break when a host moves, and why DHCP-only renumbering is not enough.
- Define home address vs care-of address and the roles of home agent, foreign agent, and collocated care-of address.
- Describe agent discovery (advertisement vs solicitation) and the agent-advertisement extension fields.
- Walk registration request/reply and the four-path data-transfer process (proxy ARP + tunneling).
- Distinguish double crossing (2X) from triangle (dog-leg) routing and the binding-cache fix plus its stale-cache problem.

Key terms

Term	Meaning in this lecture
Home address	Permanent address tying the host to its home network
Care-of address	Temporary address on the foreign network
Collocated care-of address	Care-of address when the mobile host is its own foreign agent (often via DHCP)
Home agent	Application-layer agent, usually a home-network router, that intercepts and tunnels

Term	Meaning in this lecture
Foreign agent	Application-layer agent on the visited network that delivers tunneled packets
Agent advertisement	ICMP router advertisement plus type-16 extension (care-of list on FA)
Agent solicitation	ICMP request for an agent when advertisements are missing
Proxy ARP	How the home agent impersonates the mobile host on the home LAN
Tunneling	Home agent encapsulates the original IP packet to the foreign agent
Double crossing (2X)	Remote and mobile on the same net; reply path still detours via home
Triangle / dog-leg routing	Remote → home agent → mobile instead of a direct path
Binding / warning	Cache CoA at the correspondent; invalidate it after a move

Exam lines

- IP prefixes assume a **fixed** attachment point; DHCP renumbering breaks configs, DNS, and **in-flight connections**.
- Mobile IP keeps a **permanent home address** plus a **changing care-of address**, with HA/FA (or a collocated CoA) hiding the move.
- Discovery (ICMP advertisement type **16** / solicitation) then registration (type **1**, lifetime zeros = deregister) then four-path transfer with **proxy ARP** and **IP-in-IP**.
- Transparency is complete for the correspondent — and that is why **2X** and **triangle routing** waste path length.
- Binding caches can shortcut the triangle until the host moves; then the home agent must **warn**.

M37: Cloud Computing Architecture

Learning focus

- Define cloud computing versus local storage and versus a “simple remote machine.”
- Recite the essential characteristics without which, the lecture says, a service is not a cloud.
- Contrast private, public, community, and hybrid deployment, including cloud bursting.
- Distinguish SaaS, PaaS, and IaaS (SPI) with the examples given (CRM, App Engine, Azure, AWS).
- Draw the four-layer architecture and state which layers fall under SLAs.

Key terms

Term	Meaning in this lecture
Cloud	Metaphor for Internet-based, elastic, metered remote compute — not a simple remote PC
On-demand self-service	Customer provisions without a human intermediary
Broad network access	Browser, desktop app, or mobile (e.g. iPhone apps) from anywhere
Resource pooling	Multi-tenant dynamic assignment; location opaque except region/country
Rapid elasticity	Expand and release pools as demand changes
Measured service	Metering and audited pay-per-use reports
Private / public / community / hybrid	Four deployment models
Cloud bursting	Hybrid spike handling by drawing public-cloud capacity
SaaS / PaaS / IaaS (SPI)	App / platform / infrastructure service models
Thin vs thick client	Dependent low-power vs adequate local processing
SLA	Contract whose QoS penalties apply to management and hardware layers, not the user's Internet path
OpenStack	Example stack for private-cloud create/deploy/manage

Exam lines

- Cloud is Internet access to pooled, elastic, metered resources — **hire and pay for consumption**, not a new local box.
- **All** listed essential characteristics are required, or it is **not** cloud computing.
- Choose deployment by control and community: private for mission-critical (still disaster/insider-prone), public for scale/cost (Google example), community for shared concerns (banks/traders), hybrid for bursting and mixed sensitivity.
- **SPI:** SaaS uses apps; PaaS deploys onto a vendor-managed stack (App Engine, Azure); IaaS rents raw resources (AWS). Bring-your-own package → **IaaS not PaaS**.
- Four layers from client device to data center; **network is outside SLAs**, **management and hardware are inside**, with hardware affecting penalties most.

M38: Security in Cloud Computing

Learning focus

- Define cloud computing security as procedures, processes, and standards spanning physical and logical issues in SaaS/PaaS/IaaS and public/private/hybrid delivery.
- List data-security and virtual-data-center challenges (multi-tenancy, EU privacy, disk wipe standards, loss of visibility, provider insiders, hypervisor I/O).
- Enumerate cloud **network** challenges from guaranteed bandwidth through location-dependent VLANs.

- Explain how security responsibility shifts as capabilities (and risks) are inherited up the SPI stack.
- Detail SaaS, PaaS, and IaaS (especially private-cloud) security issues named in the lecture.

Key terms

Term	Meaning in this lecture
Cloud computing security	Procedures, processes, and standards for assurance in cloud environments
Trust path	Dependency among parties in Internet applications such as e-commerce
New insider	Provider staff who can see/control customer data without being company employees
Sticky services	Provider lock-in that makes leaving (e.g. incompatible storage APIs) hard
Hypervisor	Translation layer whose I/O and security model must not be assumed safe
Broadcast-domain abstraction	Missing cloud-network feature that forces application rewrites
XML encryption / WS-Security / SSL / TLS	In-transit protections listed for SaaS
Amazon S3	Example where **data at rest are unencrypted by default**
CIA	Confidentiality, integrity, availability — QoS on virtual networks hits **availability**
Active Directory	Example traditional authentication repository that private cloud outgrows

Exam lines

- Cloud security is **physical and logical**, across **SPI** and **public/private/hybrid**, and must answer **where data live and who can see them**.
- Putting data off-premises raises **multi-tenancy, EU privacy, wipe standards, lost SIEM feeds, and provider insiders**.
- Virtual data centers hide load-time failures: hypervisor I/O, **app-unaware** VMs, huge disk images, and **portable images flooding storage networks**.
- Cloud networks struggle with **SLA bandwidth**, **appliance insertion**, **per-switch policy**, **protocol zoo**, **no broadcast domain**, **cloud IPs**, and **VLAN/port location binding**.
- **Risks inherit up the stack**; the lower the provider's security cutoff, the more the customer must implement. SaaS adds browser/transit/locality/sticky/backup issues (encrypt S3 yourself); PaaS adds object isolation and privacy; IaaS/private cloud still has **hypervisor, awkward internal multi-tenancy, IAM, and virtual-network CIA**.

- Security is required at **every interface** among virtualized hardware, infrastructure, platforms, and business apps.

M39: GSM Architecture

Learning focus

- Place GSM as the dominant 2G digital cellular standard (CEPT GSM → ETSI → 3GPP) and list GSM900 / DCS1800 / PCS1900 bands.
- Partition the network into RSS, NSS, and OSS and name the A, Abis, Um, and O interfaces.
- Identify MS hardware vs SIM identifiers (IMEI, IMSI, PIN/PUK, Ki, Kc, TMSI, LAI).
- Explain MSC, HLR, VLR, EIR, AuC, GMSC, and SMS gateways, including HLR/VLR hierarchy.
- Classify GSM teleservices, bearer services (9.6 kbps, HSCSD, GPRS), and supplementary services (CLIP/CLIR, AoC, CUG, …).

Key terms

Term	Meaning in this lecture
GSM	2G digital cellular; ~70% share; 800+ million users in 190+ countries
RSS / NSS / OSS	Radio, switching, and operation subsystems
Um / Abis / A / O	MS–BTS radio; BTS–BSC; BSS–MSC (PCM-30); management (SS7/X.25)
IMEI / IMSI	Equipment identity (in hardware) vs subscriber identity (on SIM)
PIN / PUK / Ki / Kc	Unlock SIM; unblock after 3 PIN fails; auth key; cipher key
TMSI / LAI	Temporary subscriber ID and location-area ID stored while camped
HLR / VLR	Permanent user DB vs per-MSC visitor copy (cuts long-distance signaling)
AuC / EIR	Auth/cipher secrets; IMEI white/black/grey lists (blacklists not shared across operators)
GMSC / SMS-GMSC / SMS-IWMSC	Incoming voice routing; SMS toward MS; SMS from MS
Teleservice / bearer / supplementary	Speech/SMS/fax; 9.6 kbps data path (HSCSD, GPRS); extras like CLIP and CUG
MSISDN / MSRN	Directory number vs routing number obtained from HLR

Exam lines

- GSM is Europe’s answer to fragmented analog 1G: **one digital 2G** under ETSI then **3GPP**, built to **roam** with **ISDN-like voice**, later band-extended as GSM900 / DCS1800 / PCS1900.

- Users only see **MS + BTS antennas**; the real machine is **RSS (Um/Abis) + NSS (A, HLR/VLR/MSC) + OSS (O, TMN)**.
- Identity splits: **IMEI** (stolen phone) vs **IMSI/SIM** (stolen credentials); **VLR copies HLR** so the home database is not hit on every local event.
- Incoming calls go **GMSC → HLR (MSRN) → visited MSC**; SMS has analogous **GMSC/IWMSC** pair.
- Services stack as **teleservices (13 kbps speech, SMS, fax)**, **bearers (9.6 kbps, then HSCSD/GPRS)**, and a long list of **supplementary** controls (CLIP/CLIR, barring, CUGs, AoC).
- EIR blacklists **do not sync between operators**, so a stolen handset may still work on another network.

M40: MANET - I

Learning focus

- Contrast wired networks with wireless networks and name the three wireless types the lecture uses: fixed wireless, wireless with fixed access points, and MANET.
- Define a mobile ad hoc network as an infrastructure-less, IP-based radio network in which nodes act as both hosts and routers.
- List MANET characteristics (autonomous terminals, distributed operation, multi-hop routing, dynamic topology, limited resources and security, scalability, low bandwidth).
- State the open challenges the instructor listed: scalability, routing, QoS, the client–server model shift, security, energy conservation, node cooperation, and interoperation.
- Recall the application families taught: military, home/office/education, VANET, wireless sensor networks, wireless mesh networks, disaster relief, and personal area networks.

Key terms

Term	Meaning in this lecture
MANET	Infrastructure-less, IP-based radio network of mobile nodes that route for one another
Access point	Fixed infrastructure node that mobile hosts associate with in the non-ad-hoc wireless type
Multi-hop	Relaying through intermediate nodes when source and destination are not in direct range
Dynamic topology	Unpredictable, rapid link change because nodes move freely
QoS	Guaranteed performance for a flow: delay, jitter, bandwidth, packet-loss probability
VANET	Vehicular ad hoc network; V2V between vehicles, V2I via roadside units

Term	Meaning in this lecture
WSN	Sensor-node network treated as a low-mobility special case of MANET, with sink nodes
Wireless mesh	Many 802.11a/b/g mesh routers sharing an Internet connection by dynamic hop-by-hop routing
PAN	Person-centred short-range network, including Bluetooth wire-replacement

Exam lines

- Wireless freedom of movement yields three classes; MANET is the infrastructure-less extreme.
- Every MANET node is potentially a host **and** a router; A–C communication may require B.
- Design is dominated by dynamics, scarce energy/CPU/bandwidth, and the absence of a place to hang access control.
- Routing, QoS, naming/addressing without servers, security without a CA, and merging dissimilar networks are still open.
- The same idea covers battlefields, classrooms, vehicles, sensors, city-scale meshes, disasters, and PANs.

M41: MANET – Routing Protocols

Learning focus

- Define what a routing protocol does in a MANET and why metrics and information exchange matter under changing conditions.
- Distinguish **reactive (on-demand)**, **proactive (table-driven)**, and **hybrid** routing, including the bandwidth and latency trade-offs the lecture stated.
- Explain **AODV** (RFC 3561): RREQ / RREP / RERR, expanding-ring TTL, sequence numbers, precursor lists, and the three RERR circumstances.
- Explain **OLSR** as a proactive protocol that uses **multipoint relays (MPRs)**, Hello-based neighbour sensing, MPR flooding, topology-control (TC) messages, and Dijkstra shortest paths.
- Explain **TORA** as a hybrid protocol that builds a **DAG** of **heights** using QUERY, UPDATE, and CLEAR, with route creation, five maintenance cases, and route deletion.

Key terms

Term	Meaning in this lecture
Reactive / on-demand	Build a route only when needed; higher search latency
Proactive / table-driven	Continuously maintain routes to all nodes via periodic control traffic

Term	Meaning in this lecture
Hybrid	Proactive backbone plus reactive flooding for extra demand
AODV	Ad hoc On-Demand Distance Vector (RFC 3561); RREQ, RREP, RERR
Expanding ring	Increase RREQ TTL on retransmission if no reply
Precursor list	Neighbours that likely use this node as next hop; used to send RERR
OLSR	Optimized Link State Routing; proactive, MPR-optimized link state
MPR	Multipoint relay: selected one-hop neighbour that forwards floods to two-hop neighbours
TC message	Topology-control advertisement from nodes with a non-empty MPR selector set
TORA	Temporally-Ordered Routing Algorithm; DAG of ordered heights, loop-free multiple paths
QUERY / UPDATE / CLEAR	TORA control packets for create, maintain, and delete routes

Exam lines

- MANET routing is the algorithm-as-protocol that tracks a moving graph with chosen metrics.
- Reactive protocols save bandwidth but pay **initial delay**; proactive protocols start sessions quickly but **waste bandwidth** and recover slowly from failures; hybrid needs **foreknowledge** of size and demand.
- AODV finds a path when RREQ hits the destination or a node with a valid cache, replies along reverse routes, and uses RERR plus precursor lists when links die.
- OLSR shrinks classical link state by letting only MPRs rebroadcast and by advertising only MPR-selector links; Dijkstra then yields hop-shortest routes.
- TORA orients a DAG with heights so traffic is loop-free; QUERY/UPDATE build routes, five cases repair them, and CLEAR erases a partition.

M42: 3G Network and Security

Learning focus

- Place 3G in the ITU **IMT-2000** story (spectrum, pre-commercial and commercial launches, India's MTNL/BSNL path, and the GPRS → EDGE transition).
- State what 3G is: third-generation mobile telecommunications meeting **IMT-2000**, with at least **200 kbps** and later 3.5G / 3.75G broadband rates.
- List 3G capabilities, applications, and services (mobile Internet, HTML vs plain-text email, SMS/MMS/IM plus presence).
- Sketch 3G architecture: **RAN** (Node B, RNC) and **core** (packet domain SGSN/GGSN, circuit domain MSC, CGF) over **ATM Iu/Iub/Iur** with AAL2 voice and AAL5 data.
- Recall 3G security: CIA, KASUMI vs A5/1, IMSI/TMSI and false-base-station issues, and the **ETSI TS 121 133** threat classes.

Key terms

Term	Meaning in this lecture
IMT-2000	ITU name for 3G technical specifications
EDGE / EGPRS	Enhanced GPRS step before 3G; new 3G frequencies still need new infrastructure
RAN / Node B / RNC	Radio access network; Node B ≈ 2G BTS; RNC ≈ BSC (RRM, handover)
SGSN / GGSN / MSC / CGF	Packet core (as in GPRS), circuit MSC for voice, charging gateway
Iu / Iub / Iur	ATM RAN interfaces; Iu split CS (AAL2 voice) and PS (AAL5 data)
KASUMI	3G block cipher replacing A5/1; still taught as having serious weaknesses
IMSI / TMSI	Subscriber identity; IMSI sent in clear when assigning TMSI
False BS	Rogue base station that camps the user away from legitimate paging
ETSI TS 121 133	Specification used to classify 3G confidentiality, integrity, DoS, repudiation, and service-access threats

Exam lines

- 3G is IMT-2000: ≥200 kbps (later several Mbit/s), new spectrum and infrastructure, not just a software bump from EDGE.
- The architecture splits RAN (Node B + RNC) from a dual CS/PS core, with ATM AAL2/AAL5 on Iu.
- User-facing 3G is mobile Internet, email (spam/virus fears), SMS/MMS/IM+presence, and location services — with sharp regional MMS uptake differences.
- Mutual authentication is an improvement over earlier generations, but IMSI exposure, false base stations, optional encryption, and KASUMI weaknesses remain.
- ETSI TS 121 133 organises 3G threats around CIA plus repudiation and unauthorized service access.

M43: 4G LTE

Learning focus

- Define **LTE (Long Term Evolution)** as the 3GPP 4G standard on the GSM → UMTS path, aimed at ~10× 3G speeds via new DSP and modulation, and an **IP-based** architecture with lower latency.
- Place LTE among competing 4G candidates (**UMB**, **IEEE 802.16 / WiMAX**, **802.16m / WiMAX Advanced**) and **LTE Advanced (Release 10)**.
- Separate marketing "4G" (**HSPA**) from **4G LTE**, and state OFDM, peak-rate, and VoLTE/RCS benefits as taught.
- List LTE technical objectives (throughput, latency, simplified **E-UTRAN**, mobility) and the two resources required: a capable **network** and a capable **device**.
- Recall the lecture's LTE security framing: confidentiality and integrity, plus research goals for comparing LTE cryptographic algorithms.

Key terms

Term	Meaning in this lecture
LTE	Long Term Evolution; 3GPP 4G on the GSM/UMTS path
LTE Advanced	3GPP **Release 10**, the release taught as qualifying as 4G
E-UTRAN	Simplified LTE radio architecture: no RNC, no CS domain, no DCH
HSPA	High Speed Packet Access — "4G" branding without LTE; faster 3G GSM, slower than LTE
OFDM	Orthogonal frequency-division multiplexing; why 4G is faster
VoLTE	Voice over LTE: IP voice on the LTE data network
RCS	Rich Communication Services launched from the native dialer
WiMAX / 802.16m	Competing 4G path (WirelessMAN-Advanced / WiMAX 2)
UMB	Ultra Mobile Broadband, another competing 4G standard

Exam lines

- LTE is 3GPP's IP redesign of the GSM/UMTS family: DSP/OFDM speed, simpler E-UTRAN, sub-5 ms user-plane latency in the ideal case.
- Marketing "4G" may be HSPA; **4G LTE** is taught as the pure 4G form, still needing both network and device — and the right spectrum.
- VoLTE is not only HD codecs (8 vs 13 kbps) but RCS, faster setup, better battery than Skype-class OTT, and a path to VoWiFi and cross-carrier handset interoperability.
- Security is framed as CIA plus a research programme over LTE cryptographic algorithms, not a finished protocol walkthrough in this lecture.

M44: 5G

Learning focus

- Define **5G** as fifth-generation mobile/wireless broadband as the lecture presented it, including the **IEEE 802.11ac** claim and the generation path from 1G through 4G LTE.
- State **NGMN Alliance** rate, connection-count, spectral-efficiency, coverage, signalling, and **~1 ms latency** requirements.
- Describe the taught **all-IP 5G architecture**: user terminal plus multiple simultaneous **radio access technologies (RATs)**, software-defined radios, and terminal-side choice via middleware.
- Contrast 5G with 4G on speed, capacity (~1000×), latency, millimetre-wave vs existing bands, MIMO, and small cells.
- List benefits, device/network requirements, applications (IPv6, global coverage), disadvantages, and the "nanocore / 6G" closing remarks — all as taught.

Key terms

Term	Meaning in this lecture
5G	Fifth-generation mobile/wireless; lecture also ties it to IEEE 802.11ac and a 5 GHz signal
NGMN Alliance	Source of 5G rate, massive-IoT connection, spectral, coverage, and ~1 ms latency requirements
RAT	Radio access technology; each appears as an IP link; multiple active interfaces at once
MIMO	Multiple-input multiple-output, to boost shorter-range higher-frequency 5G signals
Millimetre wave	Very high frequencies (taught as 3–300 GHz): fast, little broadcast coexistence, poor through walls — hence small cells
LTE vs WiMAX	Competing 4G; LTE became the consistent standard 5G builds on
IPv6	Addressing for mobile IP tied to network and geography in the 5G application vision
Nanocore	Speculative future 5G enhancement combined with AI

Exam lines

- 5G is taught as all-IP, multi-RAT, terminal-centric: the phone keeps several radios up and chooses a provider; networks handle mobility; vertical handovers are to be avoided.
- NGMN targets tens of thousands of users, massive sensor connections, better spectral/signalling efficiency, and about **1 ms** latency.
- Speed/capacity claims in the lecture range from 500+ Mbps typical, 1–10 Gbit/s peaks, ~1000× 4G capacity, and Huawei's "100× fastest LTE" / sub-1 ms latency — with the caveat that **no universal standard existed yet**.
- Higher bands (up to 6 GHz and mmWave) buy rate but lose range, so MIMO and dense small cells (including bounce-off-surfaces coverage) are part of the story.
- Benefits (IoT, AR, film-in-a-second, bigger data plans) sit beside unsolved security/privacy, device replacement cost, and unfinished research.

M45: VoIP Protocols

Learning focus

- Define **VoIP** as packetized digital voice/multimedia over IP instead of circuit-switched TDM, and explain why IP alone is a poor realtime transport.
- Place VoIP against the **OSI** layered model and name the three signalling/control protocols taught: **ITU-T H.323**, **IETF SIP**, and **MGCP**.
- Trace sender-to-receiver processing: codecs **G.711, G.729, G.723**, RTP/UDP/IP headers, playout buffer, and SIP/H.323 for call setup/teardown.
- List **QoS** factors: delay (source, receiver, network), jitter, packet loss, echo, and throughput.
- Recall VoIP **threat classes** (DoS, sniffing, eavesdropping, spoofing, toll fraud, SPIT) and the **defences and deployment options** the instructor listed — not attack recipes.

Key terms

Term	Meaning in this lecture
VoIP	Packet voice/multimedia over IP instead of circuit TDM
H.323	ITU-T umbrella (H.225, H.235, H.245, H.450) with RTP/RTCP over UDP and TLS setup
SIP	IETF RFC 3261 text request/response signalling; needs TLS/IPsec because proxies must see headers
MGCP	RFC 3435 gateway control; IPsec AH/ESP between media gateway and MGC
G.711 / G.729 / G.723	Example voice codecs on the send path
Playout buffer	Receiver jitter buffer; late packets discarded
SPIT	Spam over Internet Telephony — real-time voice spam
Toll fraud	Unauthorized use of paid voice service, often via signalling or configuration abuse
PSTN	Public Switched Telephone Network; still required for number translation and non-VoIP users

Exam lines

- VoIP wins on TDM cost and voice–data integration, but IP is connectionless and offers no realtime guarantees — hence codecs, RTP, jitter buffers, and signalling (H.323, SIP, MGCP).
- H.323 is a full ITU stack with H.235 authentication variants and Diffie–Hellman; SIP is lightweight and **cannot hide routing headers from proxies**, so TLS/IPsec carry the security burden; MGCP assumes IPsec to the PSTN gateway.
- QoS is delay, jitter, loss, echo (electrical vs acoustic), and throughput — not a single number.
- Threats taught are DoS, sniffing, eavesdropping, caller-ID spoofing/phishing, toll fraud, and SPIT; responses are authentication, encryption, filtering/blacklists, firewalls, physical/port security, and user caution — not protocol exploits.
- Deployment spans USB adapters, softphones, dedicated phones/routers, and wireless LAN, with scalability, PSTN interworking, and incomplete encryption as the practical limits.

M46: Introduction to DDoS

Learning focus

- Distinguish **DoS** from **DDoS** and place both as attacks on **availability** in the CIA triad.
- Explain why Internet growth and a security-light original design made flooding and related availability attacks severe.
- State attacker **motives** and organisational **consequences** as listed in the lecture.

- Name the **protocol/architecture weaknesses** the instructor blamed (stateless forwarding, lack of authentication / IP spoofing, predictable TCP handshake, mimicry of legitimate users).
- Classify DDoS **as taught**: physical (layer 1), volumetric (layers 2–4), and application-layer (layers 5–7) — class names only, not operational recipes.

Key terms

Term	Meaning in this lecture
DoS	Denial of service: prevent legitimate users from using a service (availability)
DDoS	DoS whose traffic comes from **many sources**, amplifying impact and complicating defence
Bot / botnet	Compromised machines (masters, handlers, zombies) used as a distributed attack network
IP spoofing	Forged source addresses; hides origin and defeats simple identity-by-IP defences
SYN flooding	Volumetric class that leaves TCP connections **half-open** until timeout
Volumetric attack	Layers 2–4 congestion/bandwidth exhaustion
Application-layer DDoS	Layers 5–7: low network volume, high application/request load
Ping of Death	Named vulnerability class: oversized ICMP mishandled by some OSes; patchable
CIA	Confidentiality, integrity, availability — DDoS targets availability

Exam lines

- Availability was historically under-weighted; flooding proved it can bankrupt, embarrass, or politically coerce a victim.
- DDoS is harder than DoS because of **scale**, **legitimate-looking traffic**, and **spoofed identity**.
- The Internet's **stateless forwarding**, **weak authentication**, **predictable TCP handshake**, and **easy mimicry of users** are the taught root causes — not a single bug.
- Classification is **physical vs volumetric vs application-layer**; named floods (MAC, ICMP, SYN, UDP, HTTP GET/POST) are **types**, not procedures.
- Patching stops some vulnerability DoS; **public-Internet flooding has no easy preventative**, and commercial tools have not closed the problem.

M47: Defence to DDoS Attacks

Learning focus

- Explain why **DDoS flooding** is hard to defend: easy launch, traffic similarity, IP spoofing, huge volume, many zombies, and Internet **hub-and-spoke** chokepoints.

- Separate **technical** vs **social** defence challenges (distributed response, missing incident data, no benchmarks, large-scale testing; deployment economics).
- List **goals of an ideal defence**: effectiveness, completeness, minimum collateral damage, low false positives, low cost.
- Name the four **defence modules**: prevention, detection, source identification, and reaction — as strategies, not attack methods.
- Recall the instructor's **hardening tips** (filters, patches, unused services, quotas, baselines, physical security, integrity tools, redundancy, backups, passwords).

Key terms

Term	Meaning in this lecture
IP spoofing (defence view)	Makes new legitimate clients indistinguishable from attack initiators
Collateral damage	Legitimate traffic dropped or delayed by a defence
False positive	Defence treats a non-attack (e.g. flash crowd) as DDoS
Flash event	Sudden legitimate load that must be distinguished from DoS
SYN cookies	Example partial preventive for SYN floods; not complete by itself
Anomaly-based detection	Preferred DDoS detector; signatures change too often
Source identification	Finding origins even when addresses are forged
PlanetLab / Emulab	Named research testbeds; small-scale tests are not credible
Tripwire	Example integrity tool for detecting configuration-file changes

Exam lines

- Flooding looks like a "perfect crime": cheap volume, spoofed identity, too many bots to punish, and hubs that can congest the whole Internet.
- A box next to the victim cannot keep up; real defence must be **distributed**, but **economics and reputation-secrecy** block the data, benchmarks, and deployment that research needs.
- Ideal properties are **actually stopping attacks**, covering **many attack styles**, **not punishing legitimate users** (including flash crowds), **few false alarms**, and **cost in line with benefit**.
- The operational pipeline is **prevent (especially anti-spoof near sources) → anomaly-detect → identify sources → react without collateral damage**.
- Everyday hygiene — filters, SYN-flood patches, turning off spare services, quotas, baselines, physical security, Tripwire, redundancy, backups, strong admin passwords — **reduces exposure**; it is **not** a complete DDoS solution, which the instructor says the market has not yet delivered.
